CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (EC)

December 2012

EC 407 Radar Systems

Date: 8.12.2012, Saturday | **Time: 10:00 a.m. To 01:00 p.m.** | **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q – 1	(a)	What are the main types of Navigation aids?	[03]
	(b)	Draw neat block diagram of FMCW radar and enlist its distinguished features.	[04]
Q - 2	(a)	For an N-turn loop antenna, prove that the voltages induced in all turns add up to give a resultant induced voltage $E_N=NE_f$ where E_f is o/p voltage of a loop.	[05]
	(b)	Enlist and briefly explain various errors in direction finding.	[05]
	(c)	Write major limitations of microwave landing systems.	[04]
		OR	
Q - 2	(a)	With the help of antenna patterns explain the working of a sense finder in direction finding.	[05]
	(b)	Explain Automatic direction finder through a block diagram.	[05]
	(c)	Define direction finder term and which antenna is used for this purpose and why?	[04]
Q - 3	(a)	Explain radio altimeter and draw neat waveform in support.	[05]
	(b)	Describe the principle of VOR. Derive the equation of electric field of a VOR system with five antenna configuration.	[05]
	(c)	Write short note on distance measuring equipment.	[04]
		OR	
Q - 3	(a)	Why sinusoidal modulation is used in radio altimeter? Derive an expression for average beat frequency for such modulation.	[05]
	(b)	Describe salient features of TACAN.	[05]
	(c)	Switched Cardiod Homing system.	[04]

SECTION – II

Q - 4	(a)	What is receiver noise? Derive the RADAR range equation in terms of bandwidth.	[04]
	(b)	Explain filter characteristics of delay line canceller.	[03]
Q - 5	(a)	What are the limitations of MTI radar? Explain any two in detail.	[05]
	(b)	Two MTI radars are found to have blind speeds are given by V_{b2}=20m/s of first MTI radar and V_{b3}=30m/s of second MTI radar. If they operate at a wavelength of 3 cm, then find the ratio of PRFs of the Radars.	[05]
	(c)	What do you understand by blind speed? How to eliminate it?	[04]

OR

Q - 5	(a)	Explain about staggered pulse repetition frequencies.	[05]
	(b)	If MTI radar operates at 10GHz with PRF of 0.8KHz, then find the three blind speeds.	[05]
	(c)	Describe the Non-coherent MTI radar with the help of block diagram.	[04]
Q - 6	(a)	Explain the principle of hyperbolic navigation. Compare the operation of LORAN-A and LORAN-C	[05]
	(b)	Write short note on balanced duplexer.	[05]
	(c)	Explain DECCA.	[04]

OR

Q - 6	(a)	Describe the working of OMEGA.	[05]
	(b)	Write short note on any two types of display used in RADAR.	[05]
	(c)	Explain mixers.	[04]